# XINGQIANG CHEN

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## OBJECTIVE

Looking for **FULL TIME** roles in **LLM Inference Systems AI Agent AI Compiler Applied Foundation Models Recommendation Systems**.

## INTERDISCIPLINARY BACKGROUND

Trained in theoretical physics and biophysics / condensed-matter research, spanning computational chemistry, molecular dynamics, and computer science; contest modeling and systems engineering run through academic and industry work.

**CUMCM**: **National First Prize** (2012).

**MCM (COMAP)**: **Meritorious Winner** (2013).

**Solar Decathlon**: **6th** worldwide (2013).

**IEEE ICDM Global Meteorological AI Challenge**: ranks **57** and **32** of 1739 teams (2018).

**Best Practices of Applied Algorithms**: first-round **judge** (2021).

## EDUCATION

**Ph.D. in Biophysics and Condensed Matter** Sep. 2014 - Jul. 2018
Xiamen University, Department of Physics and Engineering GPA : 3.5/4.0

**B.S. in Theoretical Physics** Sep. 2010 - Jul. 2014
Xiamen University, Department of Physics and Engineering GPA : 3.0/4.0

## SKILLS AND PROJECTS

**Programming Languages**: C++, Python; Java, Scala, Go, CUDA.

**MLOps Tools**: PyTorch, TensorFlow, Triton, TVM, Kubernetes; Hadoop, Flink, Spark.

**LLM Inference Systems**: Transformers, KV-Cache, quantization, batching & scheduling, Triton kernels, CUDA, FlashAttention, graph fusion, GPU profiling & replay.

**AI Compiler Kernel**: Graph compilation, persistent megakernel, operator fusion, cost modeling; LLM compiler & runtime (YiRage YPK); GPU trace capture/replay (TraceSmith).

**Deep Learning Models**: Knowledgeable in Transformers, RL, GNN, DNN, CNN, and more.

**Open Source Highlights**:

- **TraceSmith** — Cross-platform GPU Profiling & Replay System for AI compilers and DL frameworks (CUDA/ROCm/Metal/M lock-free capture, deterministic replay; PyPI). `https://github.com/chenxingqiang/TraceSmith`
- **Research CLI** — Command-line AI workflow tool with academic research features; multi-model providers and MCP tooling. `https://github.com/iechor-research/research-cli`
- **JAX-sklearn** — Drop-in scikit-learn replacement with automatic JAX acceleration (*13,058 tests passed; PyPI release*). `https://github.com/chenxingqiang/jax-sklearn`
- **YouTube-DNN-RecSys** — Clean TensorFlow & Spark implementations for YouTube-style recommendation models (Python+Scala). `https://github.com/chenxingqiang/YouTube-DNN-RecSys`
- **Questions-Gen** — Math competition problem generator (Qwen3-14B, RL GRPO, KD; CLI & HF models). `https://github.com/chenxingqiang/GenQuestion`

- **DeepMicroPath** — Multi-mode research agent with microbiology specialization; non-intrusive wrapper enabling auto/default/microbiology prompts; Docker/SLURM deployment. `https://github.com/chenxingqiang/DeepMicroPath`
- **Medical-NER-Qwen-4B-Thinking** — Qwen3-4B-Thinking fine-tune via LoRA for medical entity & relation extraction (English). `https://huggingface.co/xingqiang/Medical-NER-Qwen-4B-Thinking`
- **QuestionsGen-Qwen3-14B-Final (FP)** — Full-precision merged final model trained with Unsloth; deployment-ready HF weights. `https://huggingface.co/xingqiang/QuestionsGen-Qwen3-14b-final-fp-merged`
- **YiRage Persistent Kernel (YPK)** — LLM megakernel compiler & runtime achieving 1.2×–6.7× latency reduction. `https://github.com/chenxingqiang/YiRage`

## EXPERIENCE

### Senior AI Compiler Engineer, Yizhu Tech Ltd.
Hangzhou, China
***R&D Department*** *Sep. 2022 - Present*

- Led development of **YiRage Persistent Kernel (YPK)**, compiling end-to-end LLM inference into a single megakernel; on Llama2-7B achieved **1.2×–6.7×** latency reduction, **40%** higher GPU utilization, and **25%** lower memory use.
- Designed fused operators (RMSNorm+Linear, etc.), cutting **85%** of kernel-launch overhead; single-request latency from **120ms to 18ms** (batch size=1).
- Built adaptive scheduling with Triton and FlashInfer; scaled to **8 GPUs** at **15,000 tokens/s**.
- Developed a graph compiler with kernel auto-tuning and KV-cache layout; memory-access efficiency **+60%**, cache hit rate **92%**.
- Filed **2** patents; deployed to **2** datacenter customers.
- Owned ReRAM SoC compiler backend and software ecosystem: LLVM Target, ISA bring-up, and code standardization.
- Led model conversion and operator-level analysis (Cycles/MACs) across runtimes; integrated TVM, ONNX, MLCommons, TensorFlow/PyTorch into YZlite for chip benchmarks.

### Privacy-preserving Algorithms Engineer, Trustbe ( )
Hangzhou, China
***IT Group*** *Sep. 2021 - Sep. 2022*

- Built a homomorphic-encryption federated learning platform for **20+** financial institutions; **15%** higher accuracy with privacy preserved, **40%** shorter training time.
- Designed MPC protocols for joint risk models at **8** banks; fraud detection **96.7%** accuracy, **30%** lower false positives.
- Open source: **Secret-Learn**, **573** implementations (191 algorithms × FL/SS/SL), 100% sklearn-compatible, JAX **5×+** speedup, SecretFlow integration (**PyPI, 150,000+ LOC**).
  `https://github.com/chenxingqiang/Secret-Learn`
- Published **2** papers and filed **3** patents; core tech adopted by **5** products.

### Senior Algorithms Engineer, Yalla Group (NYSE: YALA)
Hangzhou, China
***Data Center Engineering Group*** *Sep. 2020 - Sep. 2021*

- Rebuilt the ChatRooms recommender with multi-objective optimization; daily active time **+35%**, 7-day retention **42% → 61%**.
- Built Ludo risk control with real-time behavior models; malicious-user detection **94%**, refund rate **−45%**.
- Developed a Texas Hold'em game AI with deep RL; **78%** win rate, play time **+25%**, paid conversion **+18%**.

### Senior Algorithms Engineer, Gejia Network Tech
Hangzhou, China
***Search & Recommendation Engineering Group*** *Mar. 2019 - Sep. 2020*

- Designed multi-scenario ranking with CF, deep models, and knowledge graphs; CTR **+28%**, CVR **+35%**.
- Optimized the real-time recommender to **1M** QPS with **<50ms** latency, **95%** coverage, satisfaction **3.2 → 4.6**.

### Machine Learning Engineer, Bianlifeng Tech
Beijing, China
***Data Engineering Group*** *Mar. 2017 - Mar. 2019*

- Built sales forecasting for **50+** stores; **92%** accuracy, inventory turnover **+30%**.
- Linked customer IDs to purchases over **10M+** events; identity accuracy **89%**, personalized sales **+22%**.
- Shipped a CV checkout model for **500+** SKUs; **96%** accuracy, checkout speed **+60%**.

## RESEARCH & INTERNSHIPS

### Molecular Dynamics Simulation of Proteins with QM/MM model
Xiamen university, China
**(China National Funds)** *Sep. 2014 - Mar. 2017*

- Conducted research on protein dynamics simulations using quantum mechanics, and analyzed biological information of proteins. This involved interdisciplinary knowledge in theoretical physics, structural biology, computational chemistry, and computer science.

**Development of Liquid Biopsy ct-DNA Kit and DNA Methylation Data Analysis** Lepu Medical, China

**(R&D Engineer Intern)** *Mar. 2016 - Mar. 2017*

- Worked on the development of liquid biopsy kits, which required specific biomarkers for cancer screening. Using existing gene chip data (TCGA&GEO), we analyzed 10,000 samples to locate the methylation site on the Septin9 human gene, an early indicator of colon cancer. This informed the development of a more accurate clinical diagnostic kit for colon cancer.

## PUBLICATIONS

- Cai Y, Su M, Ahmad A, Hu X, Sang J, Kong L, Chen X, Wang C, Shuai J, Han A. Conformational dynamics of the essential sensor histidine kinase WalK. *Acta Crystallographica Section D: Structural Biology.* 2017 Oct 1;73(10):793-803.
- Xia H, Chen X, Chen X, Cheng J, Liu Y, Chen X, Zhang Q, Zou G. Chiral discrimination of amino acid enantiomers based on different interactions with Cu2+. *Sensors and Actuators B: Chemical.* 2018 Jan 1;254:44-51.
- Ahmad A, Cai Y, Chen X, Shuai J, Han A. Conformational dynamics of response regulator RegX3 from Mycobacterium tuberculosis. *PloS one.* 2015 Jul 22;10(7):e0133389.
- Chen Y, Qi H, Li X, Cai M, Chen X, Liu W, Shuai J. Suppressing effect of C a 2+ blips on puff amplitudes by inhibiting channels to prevent recovery. *Physical Review E.* 2016 Aug 15;94(2):022411.